Sidharth Bhagat

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EDUCATION

B.Tech in Electrical Engineering | *Indian Institute Of Technology, Jodhpur(India)* Grad 2027 | CGPA: 7.88
Class XII | *Emanuel Mission School, Kota* Grad 2023 | 85%
Class X | *Techno India Group Public School, Hooghly* Grad 2021 | 95%

WORK EXPERIENCE

Software Engineer Intern – Data Engineering | *ICICI Bank, Audit Intelligence Unit* Summer - 2026 | Hyderabad

- Architected **AIU DataVerse**, a full-stack **Python** data platform with a **9-phase Bayesian delimiter detection** pipeline and **23-stage type-recovery engine** (14 numeric, 9 datetime) for automated **schema inference** on heterogeneous financial datasets.
- Re-engineered the backend from **Pandas to Polars LazyFrame** with **SHA-256 content-addressed Parquet caching**, benchmarking a **1,673x** column-profiling and **12.91x** CSV-parsing speedup via an automated regression suite.
- Built a **transformation and analytics engine** with auto-selected **regression modeling**, **MinHash + LSH** deduplication, and **PSI schema-drift detection**; independently audited the codebase, remediating **6 defects** including a critical constructor null-init bug.

Software Development Fellowship | *SoLA, Comic Studies Dept. | GitHub | LOR* Summer - 2025 | Remote

- Built and shipped an **end-to-end software pipeline** combining **computer vision**, **OCR/text extraction**, **NLP**, and **text-to-speech (TTS) APIs** to convert digital comic pages into an **accessibility-focused narrated reading experience**.
- Engineered a standardized **inference pipeline** for **panel segmentation**, dialogue extraction, **speaker attribution**, and multi-voice speech synthesis, achieving an end-to-end throughput of **4 pages/minute** across the processing workflow.
- Implemented **NLP-driven speaker attribution** for up to **5 characters** and **emotion-conditioned multi-voice TTS**, integrating extracted dialogue with character-specific narration to automate comic-to-audio conversion.

Under-Graduate Research Intern | *CS Department, IIT Jodhpur | LOR* Summer - 2024 | IIT Jodhpur

- Collaborated with **Prof. Tanmay Nitin Inamdar** on **algorithmic research** into **meta-heuristic** and approximation approaches for **combinatorial optimization**, **benchmarking heuristic algorithms** to demonstrate a **40%** improvement in solution quality for complex optimization scenarios.

PROJECTS

CIFAR-10 Image Classification with Custom CNN Architecture | *PRML Course Project | GitHub* *Feb 2025 – Apr 2025*
Technologies: Custom CNN, CUDA Acceleration, TensorFlow, Keras, Data Augmentation, ZCA Whitening, AdaBoost, XGBoost, ANN, BatchNorm

- Designed and trained an **8-layer CNN** with a **GPU-accelerated CUDA** backend, integrating **Batch Normalization**, **Dropout regularization**, and **adaptive learning-rate scheduling** to improve convergence.
- **Benchmarked** model performance against classical **SVM** and ensemble baselines (**AdaBoost**, **XGBoost**), reaching **80.5% test accuracy**, and applied **Grid Search** with **Early Stopping** for repeatable **hyperparameter tuning**.

WARDEN: Distributed Web-Change Detection & Alerting Platform | *Personal Project | GitHub* *July 2026 – Present*
Technologies: Python, FastAPI, PostgreSQL, Redis, Docker, Selenium Grid, S3-compatible Storage, OpenTelemetry, Prometheus, Grafana

- Built a **multi-tenant, event-driven monitoring platform** with JWT-authenticated APIs, Redis Streams, S3-compatible storage, SSRF isolation, and WebSocket replay; recovered **500/500 jobs across 12 worker kills** with **zero lost jobs or duplicate durable effects**.
- Reduced simulated stale time **37.7% at fixed crawl capacity** using Poisson MAP estimation and Lagrangian optimization; benchmarked **1 million timer insertions in 1.47 seconds**, with bounded-load consistent hashing and atomic Redis Lua rate limiting.
- Implemented **Myers diff, Merkle DOM hashing, SimHash, TF-IDF, and SSIM**; added tenant-specific **online logistic regression**, shadow scoring, and gated promotion, improving **F1 from 0.671 to 1.000** on 600 synthetic held-out examples.
- Built a **typed rule DSL** using a handwritten Pratt parser and RE2, validated through **60,000 fuzz cases**; implemented transactional-outbox alerts, HMAC-SHA256 webhooks, email, cooldowns, retries, circuit breakers, and dead-letter queues.
- Reduced PostgreSQL query latency by **up to 99.6%** on 150,000-row test data using indexes and **EXPLAIN ANALYZE**; instrumented OpenTelemetry context propagation, structured logs, Prometheus SLO metrics, and Grafana dashboards.
- **Roadmap:** React/TypeScript dashboard, Django/DRF analytics, PostgreSQL streaming read replicas, PgBouncer transaction pooling, and Kubernetes deployment.

For more projects, visit: Sidharth8100.github.io

SKILLS

- **Languages:** Python, C++, SQL, JavaScript, TypeScript, HTML/CSS
- **Systems, Backend & APIs:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Multithreading & Concurrency, Distributed Systems, Microservices, System Design, FastAPI, Django, Django REST Framework (DRF), RESTful APIs, React
- **Data Engineering & Databases:** Polars, Pandas, NumPy, PostgreSQL, Redis, Parquet, ETL Pipelines, Schema Inference
- **Machine Learning & Computer Vision:** PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, CUDA, CNNs, XGBoost, NLP, OCR
- **Cloud, DevOps & Tools:** Docker, Kubernetes, Git, GitHub, Linux (Ubuntu), CI/CD, Prometheus, Grafana, Google Cloud APIs, Agile/Scrum

ACHIEVEMENTS

- **Guardian (top 1.02%)** in LeetCode | 5 stars in codeChef (800+ Questions, Exceptional in problem solving)
- **Departmental Rank: 3** in program B.Tech at the end of First Year.
- Secured an **A- (Excellent)** in **Linear Algebra and Differential Equations** and **Differential Integral Calculus, Computer Vision**.
- Ranked in Top **1.5 %** of 1.2M+ candidates in the JEE (Joint Entrance Examination).

POSITIONS OF RESPONSIBILITY

- Assistant Head - Varchas'25 | Prometeo'25 (Technical Fest), IIT Jodhpur.